

Transverse optical torque observed at the nanoscale

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Ryoma Fukuhara¹, Tsutomu Shimura¹ & Yoshito Y. Tanaka^{1,2}  

Optical forces and torques acting on resonant nanostructures smaller than the wavelength of light have attracted interest in nanoscience and nanotechnology. However, experimental characterization at the nanoscale remains challenging due to the diffraction limit of light. Here we present an approach for the three-dimensional measurement of nanoscale optical forces and torques. This is achieved through the optical trapping and precision spatial tracking of a designed microscale structure that contains embedded target nanostructures. Our method enables the confinement and measurement of nanostructure positions and orientations across three translational and three rotational degrees of freedom, independent of the size, shape and material of the nanostructure. Using this method, we observe transverse optical torque on plasmon-resonant nanostructures and reveal that this behaviour is governed by the optical helicity rather than the angular momentum of incident light. This versatile platform advances our fundamental understanding of nano-optomechanical interactions and opens up possibilities for precise optical manipulation and nanoactuator design.

The intrinsic nature of light as a carrier of linear and angular momentum manifests in its ability to exert forces and torques when interacting with matter. Although these optical phenomena remain generally negligible for macroscopic bodies, they become dominant at microscopic scales and have attracted considerable attention due to their successful applications across diverse fields, including biophysics^{1,2}, microfluidics³ and micromechanics⁴. Recently, there has been increased interest in applying optical forces and torques to nanoscale objects smaller than the wavelength of light⁵, highlighting the potential for advances in nanoscience and nanotechnology, including nanofabrication⁶, nanometrology⁷ and high-sensitivity sensing^{8,9}. However, working at the nanoscale is challenging because of the diffraction limit of light. As the interaction between light and nanoparticles decreases as the particle volume decreases, the optical force and torque on particles become smaller than the effect of thermal fluctuations.

Resonant interactions between light and particles have been used to amplify the optical force and torque acting on nanoparticles^{10,11}. The resonant optical force and torque magnitudes strongly depend on the incident light wavelength, reaching a maximum near the resonance peak wavelength. Additionally, the wavelength dependence of the

optical force and torque allows the selective optical manipulation and sorting of nanoparticles that exhibit a resonant optical response¹². In the nonlinear optical regime, not only the magnitude but also the direction of the resonant forces can be changed via stimulated emission^{13,14}, enhancing selective optical sorting¹⁵. Furthermore, precisely designed metal nanostructures can control the linear and angular momentum of scattered light by engineering their supported localized surface plasmon resonances (LSPRs), that is, the collective oscillations of free electrons^{11,16,17}. This allows the nanostructures to experience optical forces and torques independent of the incident light momentum. Therefore, unlike conventional optical actuators^{3,4,18}, plasmonic actuators consisting of nanostructures that generate optical forces and torques eliminate the need for oblique incidence, shaping and steering of the light beam for incident momentum control^{19–21}.

Light diffraction also restricts the degrees of freedom of the rotational axis of optical torque acting on nanoparticles. For particles larger than the wavelength of interacting light, angular momentum based on non-uniform light irradiation that generates a biased optical pushing force can generate an optical torque around an axis perpendicular to the optical axis, that is, a transverse optical torque²². However, such

¹Institute of Industrial Science, University of Tokyo, Tokyo, Japan. ²Laboratory of Nanosystem Optical Manipulation, Department of Photonics and Optical Science, Research Institute for Electronic Science, Hokkaido University, Sapporo, Japan. ✉ e-mail: ytanaka@es.hokudai.ac.jp

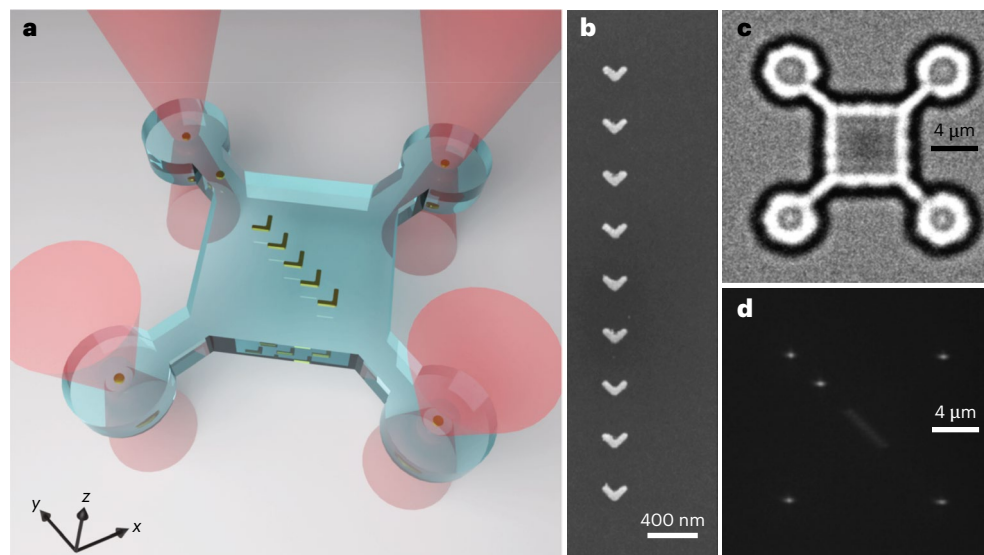


Fig. 1 | Micro-platform for measuring the 3D optical force and torque acting on nanostructures. **a**, Schematic of an optically actuated micro-platform, which consists of a central body with an embedded target structure and four arms for optical trapping and measuring its position and orientation in three translational and three rotational directions, respectively. Four laser beams, tightly focused on the arms (indicated by pink regions), enable six-degrees-of-freedom optical trapping of the micro-platform. **b**, Scanning electron microscopy image of V-shaped nanostructures with a 400-nm period embedded in the micro-platform. **c,d**, Bright-field (**c**) and dark-field (**d**) microscopic images

of the optically trapped micro-platform without and with index matching of the micro-platform and solvent, respectively (Supplementary Videos 1 and 2). The bright spots at the four corners in **d** represent the 100-nm-diameter embedded discs that allow to optically trap the micro-platform and measure its position and orientation. Another bright spot in the upper-left arm was used to roughly identify the orientation of the V structures. Index matching prevents optical force and torque generation due to light reflection and refraction on the micro-platform surface, improving the signal-to-noise ratio in the measurement of the optical force and torque acting on the target structures.

an approach is not applicable to nanoparticles that receive uniform light irradiation according to dipole approximation. Since the angular momentum of the propagating plane wave with circular polarization²³ and vortex beam with phase distributions²⁴ is directed along the optical axis, only the optical torque around the optical axis can be exerted on the nanoparticle. To address this limitation, transverse angular momentum in structured light, such as evanescent waves, two-wave interference fields, focused fields and near-field of nanostructures^{25–30}, has been studied. Furthermore, our recent work has theoretically demonstrated that a designed metal nanostructure experiences transverse optical torque under simple plane-wave illumination¹⁷. This torque arises from LSPR-induced angular momentum that is based on optical pulling and pushing forces³¹ on coupled dipoles, providing new degrees of freedom for plasmonic nanoactuators.

However, optical forces and torques on nanoparticles have not been well characterized experimentally, because they are difficult to measure owing to the diffraction limit of light. The most common approach is to optically trap the target objects in the translation and rotation directions and precisely measure their position and orientation fluctuations to determine the optical forces and torques acting on them^{32–34}. Advancements in single-molecule tracking techniques have allowed precise nanoparticle position measurements^{35,36}. However, these techniques are strongly dependent on the nanoparticle material used. Furthermore, the optical confinement and orientation measurements of nanoparticles in the three rotational directions, especially in transverse rotational directions, remain a major challenge owing to the diffraction limit²⁹.

In this paper, we present an approach that enables the three-dimensional (3D) characterization of optical forces and torques acting on nanoscale particles. Our approach circumvents the diffraction limit through optical trapping and spatial tracking of a designed microscale structure in which the target nanoparticles are embedded. This versatile method is applicable to nanoparticles of a wide range of sizes, shapes and material compositions, as it eliminates the need for their direct optical trapping and spatial measurements. By immobilizing nanoparticles within an optically trapped microstructure that is

controlled in both translational and rotational degrees of freedom, we can observe their intrinsic anisotropic mechanical responses that would typically be obscured by thermal orientation fluctuations. Using this method, we demonstrate the experimental observation of plasmon-induced transverse optical torque acting on anisotropic nanostructures and reveal its unique switching behaviour governed by the optical helicity rather than the angular momentum of incident light.

Basic concepts of nanoscale optical force and torque measurements

Figure 1a shows the schematic of an optically actuated micromachine that characterizes the optical forces and torques acting on nanostructures, which we call a micro-platform. The micro-platform is a silicon dioxide microstructure that consists of a central body and four arms (Supplementary Fig. 1). The target nanostructures (Fig. 1b) are embedded in the central body, which suppresses their Brownian motion and preserves their configuration in the solvent environment³⁷. Four laser beams tightly focused on the micro-platform arms allow the optical trapping of the micro-platform along three translational and three rotational axes. A similar approach was used to optically control the position and orientation of microprobes in nanoimaging^{38,39}. The micro-platform position (x, y, z) and orientation angle ($\theta_x, \theta_y, \theta_z$) under trapping potential $U(l)$, where $l = x, y, z, \theta_x, \theta_y, \theta_z$, can be tracked through the 3D motion of the four arms. In general, the position and orientation of particles transition stochastically following the Boltzmann distribution at thermal equilibrium^{32,34}, that is, the probability density $\rho(l) = \exp(-U(l)/k_B T)/Z$, where Z is the partition function normalizing the probability density, k_B is the Boltzmann constant and T is the absolute temperature. Therefore, trapping potential $U(l)$ can be estimated via a statistical analysis of the measured position and orientation distributions of the optically trapped micro-platform.

When another laser beam is irradiated onto the target nanostructures in the central body, the optical forces and torques generated on the nanostructures act on the optically trapped micro-platform, leading to changes in the trapping potentials along the three translational

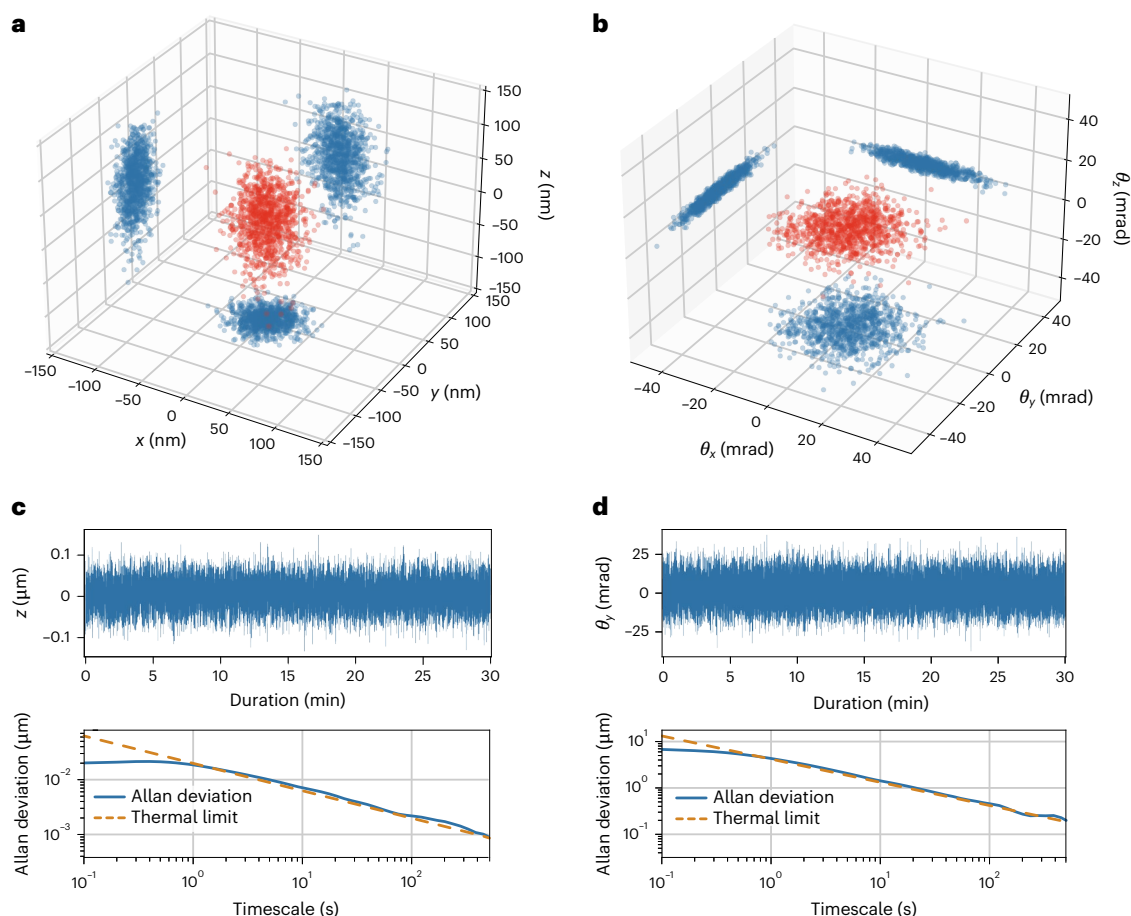


Fig. 2 | Measured position and orientation angles of the optically trapped micro-platform. a, b, Scatter plots showing positions in the x, y and z directions and orientation angles around the x, y and z axes, respectively (Methods provides the tracking details). The red and blue points indicate the 3D distributions and their projections onto each orthogonal plane, respectively. The micro-platform exhibits a larger position distribution in the z direction and angular distributions around the x and y axes, corresponding to lower trap stiffnesses compared

with other axes (Table 1). **c, d,** Time-dependent trajectories of the z position and the orientation angle around the y axis over time, respectively, and their Allan deviation analyses. The orange dashed line in the bottom panel represents the fundamental thermal limit that cannot be exceeded by any measurement. These Allan deviations show that the measurement precision reaches the thermal limit for data collection periods up to 100 s, beyond which the system drift noise becomes appreciable.

and three rotational axes. Therefore, by analysing the potential changes in the micro-platform, the 3D forces and torques on the nanostructures can be obtained. In particular, the optical trapping and measurements of position and orientation of the micro-platform can be designed independent of the size, shape and material composition of the target nanostructure. This eliminates the conventional requirement that target objects must be optically trapped and strongly emit scattered light or fluorescence to measure their position and orientation. Thus, the scope of measurement targets extends from nanoparticles with a low refractive index and low fluorescence yield to complex-shaped nanostructures, such as asymmetric nanoparticles and nanoparticle arrays^{11,16,17,19–21} (Fig. 1b).

Micro-platform sensor

The micro-platforms shown in Fig. 1 were fabricated through top-down processes on a glass substrate covered with a Si sacrificial layer (Methods). After removing the sacrificial layer via gas etching, the micro-platforms remaining on the substrate were dispersed in dimethyl sulfoxide liquid. Dimethyl sulfoxide is highly stable at room temperature and is transparent in the visible and near-infrared regions. Furthermore, its refractive index (1.47) is close to that of silicon dioxide, which constitutes a micro-platform. This prevents optical force and torque generation due to light reflection and refraction on the micro-platform

surface (Fig. 1c,d), improving the signal-to-noise ratio in the measurements of optical force and torque acting on the target nanostructures. Using gold nanodiscs embedded in the micro-platform arms, the micro-platform was optically trapped and its position and orientation were measured under dark-field illumination (Supplementary Video 2).

Figure 2a,b shows the 3D scatter plots of the centre positions and orientation angles of the optically trapped micro-platform. The astigmatic particle-tracking method^{35,36} was used for these measurements. By inserting a cylindrical lens into the optical path of the imaging system, the aspect ratio of the scattered spots from the nanoparticles in the micro-platform arms (Fig. 1d) changed with the z position along the optical axis (Supplementary Figs. 2 and 3). The 3D positional information of the four nanoparticles allows the precise analysis of the micro-platform position and orientation in the three translational and three rotational directions, respectively (a detailed explanation is provided in Supplementary Section 1), resulting in the corresponding trapping potentials (Supplementary Fig. 4). Table 1 summarizes the mean values and standard errors of the translational and rotational trap stiffnesses of the trapping potentials calculated from ten repeated measurements. These results indicate that the optically trapped micro-platform has a larger position distribution along the z axis and larger angle distributions around the x and y axes, and the corresponding trap stiffnesses are smaller than those of the other axes.

Table 1 | Translational and rotational trap stiffnesses along the x, y and z axes

Axis	Translational stiffness (fN nm ⁻¹)	Rotational stiffness (fN μm mrad ⁻¹)
x	14.5 ± 0.3	42.6 ± 1.2
y	15.3 ± 0.2	35.9 ± 1.9
z	2.95 ± 0.08	1,070 ± 15

Values represent the mean of ten independent measurements with the standard error of the mean.

Table 2 | Minimum errors in the micro-platform position and angle measurements

Axis	Position measurement error (nm)	Force measurement precision (fN)	Angle measurement error (mrad)	Torque measurement precision (fN μm)
x	1.1	16	0.38	16
y	1.1	17	0.45	16
z	2.2	6.5	0.045	48

Values correspond to the Allan deviations at a data collection time of 100 s (Fig. 2c,d), and the corresponding force and torque measurement precisions estimated from these errors and the stiffnesses values listed in Table 1.

Let us discuss the optical force and torque measurement precision by quantifying noise in our optical trapping system using Allan deviations^{40,41}. In general potential analysis, the optical force and torque can be obtained by multiplying the translational and rotational trap stiffnesses by the potential shifts induced along the corresponding axes. Allan deviations indicate the standard error of the averaged potential position over a certain timescale that is long compared with the autocorrelation time of the trap. As shown in Fig. 2c,d, the precision achieves a thermal limit that could not be exceeded by any measurement, and it improves with the averaging time. However, the drift noise of the system cannot be ignored for data collection times longer than 100 s. Therefore, the optical force and torque measurement precision should be maximized at a data collection time of 100 s, which is estimated to be 6.5 fN in the z direction and 16 fN μm along the x and y axes (Table 2). It should be noted that in contrast to the position and angle measurements, the weaker the trap, that is, the smaller the stiffness (Table 1), the more precise is the optical force and torque measurement.

Nanostructure resonance characterization via optical force measurements

To demonstrate the impact of our approach, we characterized the optical force and torque acting on anisotropic V-shaped nanostructures (Fig. 1b), which are challenging to measure using conventional methods. In a previous work¹⁷, we theoretically revealed that the interaction between the V nanostructure and circularly polarized plane-wave light via its plasmon resonance modes can generate an extraordinary transverse torque perpendicular to the optical axis, that is, the angular momentum axis of the incident light. However, there have been no effective methods for observing this nanotorque due to the diffraction limit. To experimentally demonstrate the transverse torque phenomenon, we first characterized the plasmon resonance of the nanostructures using optical force measurements. The magnitude of the optical force **F** pushing the structure along the propagation direction of the incident light beam is proportional to the extinction cross-section σ_{ext} of the structure, that is, the plasmon resonance, as follows:

$$\mathbf{F} = \frac{\sigma_{\text{ext}}}{c} \mathbf{S}, \quad (1)$$

where **S** is a pointing vector along the light propagation direction and *c* is the speed of light. The measured optical force in the z direction,

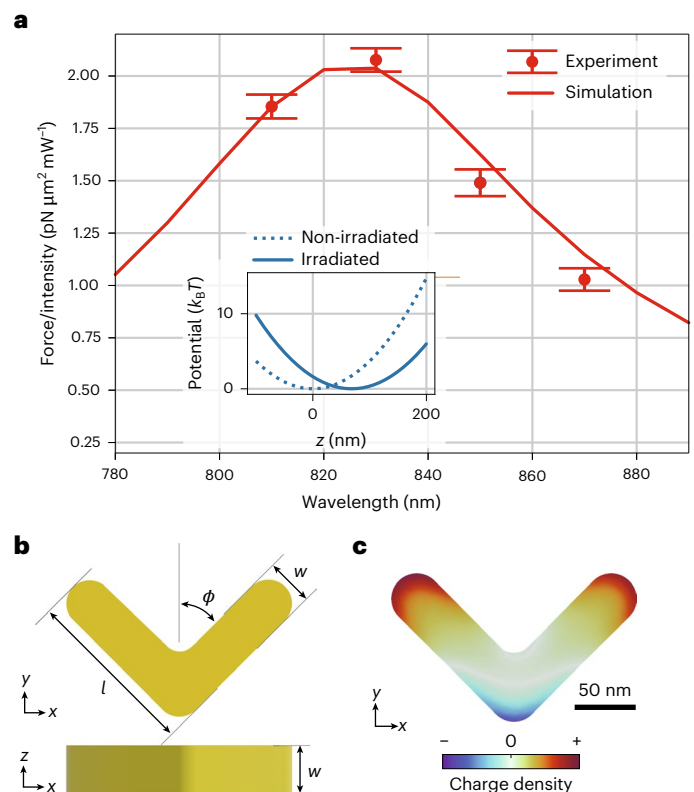


Fig. 3 | Plasmon resonance of the V-shaped structures characterized by optical force measurements. **a**, Measured optical forces (circles) in the z direction at different incident wavelengths, exhibiting a peak at a wavelength of approximately 830 nm. Each data point represents the mean of three force measurements, with error bars indicating the standard error of the mean. When the structures embedded in the central body of the micro-platform shown in Fig. 1 were irradiated with a circularly polarized light beam propagating in the positive direction along the z axis, a positive shift in the trapping potential in the z direction was observed, as shown in the inset (Supplementary Figs. 7a and 8a show the data on all six degrees of freedom with and without the V-shaped structures, respectively). The optical force was obtained from this shift and the trap stiffness listed in Table 1. **b**, Simulation model design based on the scanning electron microscopy image of the V-shaped nanostructures shown in Fig. 1b (*l* = 145 nm, *w* = 40 nm, ϕ = 45°) (Methods provides the simulation details). The solid curve in **a** shows the simulation results (scaling factor of 1.1), which corroborates the experimental result. The discrepancy between the experimental and simulation results can be attributed to differences between the actual nanostructure (Fig. 1b) and the simplified simulation model shown in **b**, particularly regarding the structure height and its profile. **c**, Calculated charge density distribution on the top surface of the representative V-shaped nanostructure for the corresponding plasmon resonance mode. This represents the symmetric mode that can induce a transverse torque around the y axis¹⁷.

that is, the incident light direction, depends on the incident wavelength and reaches a maximum at approximately 830 nm, which is well supported by the simulation results (Methods provides the calculation method; Fig. 3a). The difference between the experimental and simulation results is smaller than in previous related studies^{19,42} and can be mainly attributed to differences between the actual nanostructure (Fig. 1b) and the simplified simulation model (Fig. 3b). The observed wavelength dependence corresponds to a plasmon resonance of the symmetric mode (Fig. 3c), which should induce a transverse torque around the y axis¹⁷. In particular, spectral measurements of the optical force in the z direction yielded sufficient signal-to-noise ratios even for a single V-shaped nanostructure (Supplementary Fig. 5), showing the effectiveness of our approach for characterizing the resonance spectra of nanostructures. Although electromagnetic interactions among the

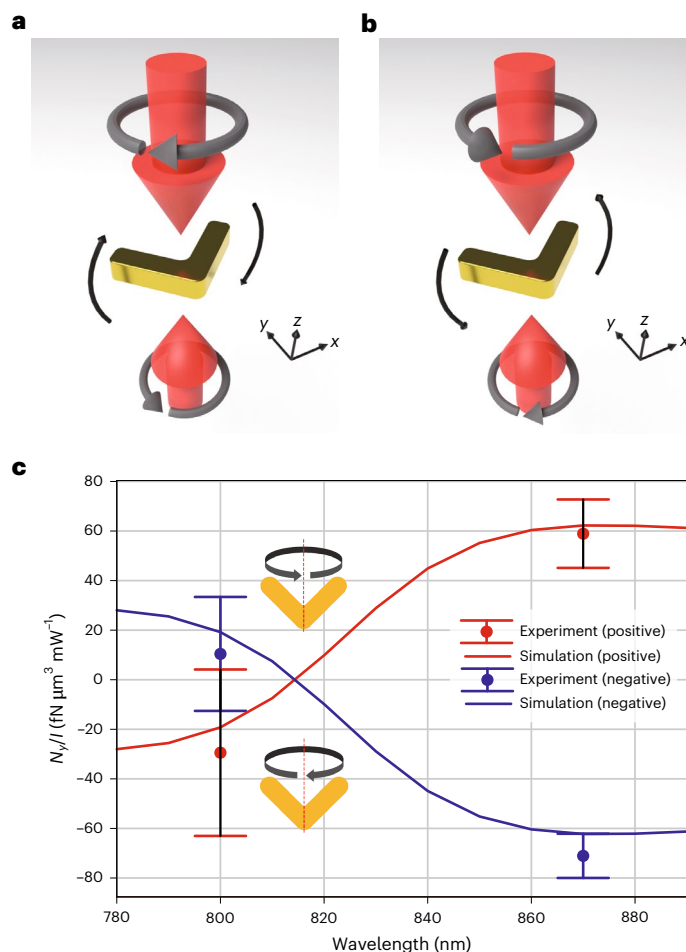


Fig. 4 | Plasmon-induced transverse optical torque on the V-shaped structures. **a,b**, Irradiation of the nanostructures with a pair of counterpropagating circularly polarized light beams (red arrows) with different spin angular momentum signs but the same optical chirality sign. The spin angular momentum of the incident light is cancelled whereas positive **(a)** and negative **(b)** optical chirality are retained. The shifts in the trapping potential around the *y* axis were observed for the nanostructure irradiation of **(a)** and **(b)** (Supplementary Fig. 7b,c shows data on all six degrees of freedom with the structures, and Supplementary Fig. 8b shows the data without the structures). The potential shift direction, that is, the transverse torque direction along the *y* axis, varied with the incident optical chirality sign. The incident wavelength was 870 nm. **c**, Measured and simulated transverse torques at different incident wavelengths. The insets display the corresponding direction of the transverse torque. Each data point represents the mean of ten torque measurements, with error bars indicating the standard error of the mean. For both cases in **a** and **b**, the measured transverse torque direction is reversed at incident wavelengths shorter (800 nm) and longer (870 nm) than the plasmon resonance peak at 830 nm in Fig. 3a. The simulation results are scaled by a factor of 1.2.

arrayed nanostructures induced spectral shifts⁴³, the relatively large inter-structure spacing (400-nm period) in this array limits near-field coupling effects, keeping the symmetric mode excitation dominant (Supplementary Section 2 and Supplementary Fig. 6).

Transverse optical torque observation

As shown in the previous theoretical study¹⁷, the direction of the transverse optical torque acting on a plasmonic V-shaped nanostructure can be changed by altering the circular polarization rotation direction of the incident plane-wave light, which is expected to provide unprecedented control of optical nanoactuators^{19–21}. To clarify this intriguing phenomenon, let us consider the physics behind it. The transverse

torque acting on the symmetric mode in Fig. 3c under circularly polarized light illumination is given by¹⁷

$$N_y = -\frac{l}{4}\alpha'_s \left(\text{Re} \left[E_y^* \frac{\partial E_x}{\partial z} \right] \right), \quad (2)$$

where α'_s is the real part of the polarizability of the symmetric mode (Methods); E_x and E_y are the *x* and *y* electric field components of the incident plane-wave light, respectively; and * denotes the conjugation operation. Since the *z* component and transverse derivatives of the incident plane-wave light are zero and $E_x^* \partial E_y / \partial z = -E_y^* \partial E_x / \partial z$ for circularly polarized light, equation (2) can be rewritten as

$$N_y = -\frac{l}{8}\alpha'_s \text{Re} [\mathbf{E}^* \cdot \nabla \times \mathbf{E}] = -\frac{l\omega}{8}\alpha'_s \text{Im} [\mathbf{E}^* \cdot \mathbf{B}] \quad (3)$$

in a monochromatic field at angular frequency ω . The circularly polarized light possesses both spin angular momentum and helicity, which are closely related to each other but exhibit opposite symmetry with respect to time reversal and spatial inversion. The optical chirality^{44,45}, based on the Lipkin's 'zilch', is a rediscovered measure of optical helicity and its time-averaged density in a monochromatic field is defined as

$$C = -\frac{\omega\varepsilon}{2} \text{Im} [\mathbf{E}^* \cdot \mathbf{B}], \quad (4)$$

where ε is the electric permittivity of a lossless medium. It should be noted that even in the absence of the spin angular momentum of light, the optical chirality in equation (4) is not necessarily zero, and vice versa. By substituting equation (4) into equation (3), we obtain

$$N_y = \frac{l}{4\varepsilon}\alpha'_s C. \quad (5)$$

Equation (5) indicates that the helicity of incident light, rather than its spin angular momentum, is fundamentally crucial for generating the transverse torque acting on the V structure. This is an insight that has not been considered in a previous theoretical study¹⁷. To clearly identify the origin of the transverse torque phenomenon, a pair of counterpropagating circularly polarized light beams with different spin angular momentum signs but the same optical chirality sign were irradiated onto the V-shaped structures in the optically trapped micro-platform (Fig. 4a,b). In this irradiation system, the spin angular momentum of the incident light was cancelled and the optical chirality was maintained. Additionally, no standing waves were generated because the optical path difference between the two laser beams was sufficiently larger than the coherence length of the laser used. When the V-shaped structures were irradiated with the counterpropagating beams, a shift in the trapping potential around the *y* axis of the micro-platform, that is, a transverse torque, was observed (Fig. 4c; also see Supplementary Fig. 7b,c). The direction of shift changed with the incident optical chirality sign. Note that this observation was obtained for multiple V structures arranged (Fig. 1b), whereas the shift was negligibly small for a single V structure. A shift in the trapping potential along the *z* axis, that is, the pushing optical force, was negligible in the irradiation system, confirming that the two counterpropagating beams had the almost same intensity on the nanostructures. These results demonstrate that the observed transverse torque is induced by the helicity of incident light, rather than by the incident spin angular momentum. Furthermore, equation (5) indicates that the transverse torque also depends on the real part of polarizability of the symmetric mode, α'_s , which changes sign across the plasmon resonance. Indeed, for both positive and negative optical chiralities, the observed torque signs differed at wavelengths shorter (800 nm) and

longer (870 nm) than the resonance peak of the symmetric mode at 830 nm (Fig. 3), as shown in Fig. 4c. Additionally, the torque magnitude at 870 nm was larger than that at 800 nm, which is well supported by the simulation results (the calculation method is provided in the Methods). Thus, our optical force and torque characterization using an optically actuated micromachine clearly reveals that the transverse torque acting on the V-shaped nanostructures originates from the interaction between their plasmon mode and the helicity of incident light.

Discussion and outlook

The independence of our methodology from the direct optical trapping of target nanostructures relaxes the constraints on their size, shape and material composition. Our demonstration using multiple immobilized V-shaped structures confirms that the total signal of optical forces and torques can be strategically accumulated on a single micro-platform sensor, thereby facilitating the characterization of structures with inherently small optical cross-sections. Furthermore, by decoupling the tracking signals from the target's optical properties, the measurable scope extends to nanostructures with low fluorescence yields or those composed of complex, non-transparent materials.

Conventional optical tweezer-based methods have been limited to single-axis rotational confinement and measurement³³ of nanostructures, making the full vectorial characterization of anisotropic optical forces and torques experimentally inaccessible. Our approach enables the all-axis control and read-out of both translational and rotational degrees of freedom, allowing direct experimental access to and rigorous characterization of phenomena such as negative optical force and torque^{46–49}, nonlinear resonant optical force and torque^{13,14,50}, and metamaterial-induced force and torque^{16,51–53}. This enables precise optical nanomanipulation and nanoactuator designs, such as selective resonant optical sorting^{12,15} and self-stabilizing metamaterial actuators⁵³.

Beyond the applications in optical manipulation, our methodology provides a versatile platform for the 3D measurement of femtonewton-order forces and femtonewton–micrometre-order torques, poised to advance the frontiers in fields ranging from biophysics to quantum physics by enabling the experimental investigation of mechanical interactions previously constrained by limited measurement capabilities. In single-molecule biophysics², our method enables a complete vectorial characterization of molecular mechanics, moving beyond the single-axis torque detection³³ achievable with conventional optical tweezers by resolving hitherto-inaccessible transverse components. This permits the rigorous analysis of torque-generating molecular machines such as rotary motors⁵⁴, RNA polymerase² and the RuvAB complex active in DNA recombination⁵⁵. Concurrently, our technique advances experimental studies of the Casimir effect—arising from quantum electromagnetic fluctuations—beyond traditional force measurements⁵⁶ between metallic spheres and plates. Full six-degrees-of-freedom optical trapping of complex structures enables direct access to previously theoretical forces and torques, such as repulsive interactions on chiral nanostructures⁵⁷ and torques on lamellar gratings⁵⁸. Furthermore, this trapping scheme offers an ideal platform for studying quantum optomechanics by implementing a key theoretical proposal^{59,60}: optically suspending a disc mirror that serves as one end of a Fabry–Pérot cavity, thereby decoupling thermal noise and enhancing the mechanical quality factor.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41567-026-03268-6>.

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Methods

Micro-platform fabrication

A total of 1,600 micro-platforms were fabricated for each experiment using a combination of electron-beam lithography and dry-etching techniques¹⁹. The overall fabrication process is summarized as follows. (1) A 300-nm-thick silicon layer as a sacrificial layer, followed by a 400-nm-thick SiO₂ layer were sputter deposited on a coverslip substrate (Supplementary Fig. 9a). (2) A positive resist film for electron-beam lithography was spin coated onto the substrate with a thickness of 200 nm. This resist film was exposed to a specific nanostructure pattern using an electron-beam lithography system (100 kV) and subsequently developed. A 5-nm titanium layer was deposited as an adhesion layer via electron-beam evaporation, followed by a 40-nm gold layer. The resist pattern was then removed from the substrate, resulting in the fabrication of V-shaped Au nanostructures and five nanodiscs on the SiO₂ layer (Supplementary Fig. 9b). (3) A further 400-nm SiO₂ layer was deposited, and the nanostructures were sandwiched between two SiO₂ layers (Supplementary Fig. 9c). (4) The sample was then coated with a resist to define the micro-platforms during the second electron-beam lithography process. A 100-nm-thick aluminium layer acted as a mask for the micro-platforms to etch SiO₂ with CHF₃, which was subsequently removed using an aluminium etchant (Supplementary Fig. 9d–f). (5) In the final stage of the process, the sacrificial Si layer was removed via dry etching with XeF₂ to release the micro-platforms from the substrate (Supplementary Fig. 9f).

Measurement system with a micro-platform sensor

Supplementary Fig. 10 illustrates an optical measurement system. Approximately 1% (10–20 pieces) of the micro-platforms on a coverslip surface were spontaneously dispersed in dimethyl sulfoxide sandwiched between two coverslips separated by a 50- μ m-thick spacer and settled on the lower coverslip owing to the gravitational force. Mild ultrasonication (3 s, 120 kHz) increased the dispersion yield to over 70%. The sample chamber was sealed to prevent evaporation and convection of the solvent. The holographic optical tweezer technique³⁸ using a continuous-wave neodymium-doped yttrium aluminium garnet laser (wavelength, 1,064 nm; Laser Quantum, Mod. Opus 1064) was used for the optical trapping of the micro-platform. The holographic optical tweezer system consisted of an oil-immersion objective (numerical aperture, 1.49; Nikon CFI Plan Apo TIRF), a spatial light modulator (Hamamatsu, LCOS-SLM X10468-07) and 4f optics, and produced four focused laser beams using first-order diffracted light from the LCOS-SLM. Zeroth-order light was spatially filtered at the focal plane of the 4f optical path to avoid irradiation on the V-shaped nanostructures at the centre of the optically trapped micro-platform. The total input laser power was 14.0 mW (3.50 mW per spot). The focused spot diameter was measured to be ~ 1 μ m, corresponding to an intensity of ~ 4.5 mW μ m⁻² per spot. To minimize the influence of non-optical interactions from the chamber boundaries (a detailed explanation is provided in Supplementary Section 3), the laser beam was focused ~ 25 μ m away from both upper and lower coverslip surfaces, that is, at the centre of the sample chamber. A continuous-wave Ti:sapphire laser (Spectra-Physics 3900S) with a tunable wavelength between 800 and 880 nm in our system was used to generate optical forces and torques on the V-shaped structures. The laser beam was split into two beams and counter-propagated within the sample chamber. The two laser beams propagating in the positive and negative *z* directions were gently focused through the same objective as that of the trapping laser beam and another aspheric lens (Thorlabs, C240TMEB). The diameters of the incident laser beams were controlled to achieve the same focused numerical aperture (<0.5). The input laser powers were 0.45 mW and 1.60 mW for the experiments shown in Figs. 3 and 4, respectively. With a measured spot diameter of ~ 1.7 μ m, the corresponding intensities were approximately 0.2 and 0.7 mW μ m⁻². The effect of

this laser irradiation on the trap stiffness was negligible. Circular polarization on the nanostructures was formed and controlled using both half- and quarter-wave plates (Thorlabs, SAHWP05M-700 and SAQWP05M-700). To measure the position and orientation of the optically trapped micro-platforms under dark-field illumination, a pulsed white light-emitting diode (Sawaki, FOLS-10) was introduced through the same optical path as that of the trapping laser beam. The light-emitting diode was synchronized with the imaging complementary metal–oxide–semiconductor sensors (Andor, Zyla 4.2 Plus) to prevent the rolling-shutter phenomenon. Furthermore, to obtain the *z* positions of the scatterers embedded in the micro-platform arms, a cylindrical lens (Thorlabs, LJ1516RM-A) was incorporated into the imaging system and astigmatism was introduced into the point-spread function of their scatterers, transforming it into an ellipse (Supplementary Fig. 2a). From the distributions of the positions and orientations of the optically trapped micro-platforms, the centre positions and spring constants of the trapping potentials in three translational and three rotational directions were estimated using a maximum likelihood method⁶¹ (a detailed explanation is provided in Supplementary Section 4).

Micro-platform tracking

The centre positions and orientation angles of the optically trapped micro-platform were tracked by analysing the 3D positional data of four scatterers embedded in its arms. The platform motion was recorded for 100 s at 20 Hz, well above the optical trap's corner frequency ($f_c < 3$ Hz from the power spectrum analysis³⁴), ensuring sufficient temporal resolution and preventing aliasing artifacts⁴⁰. Motion blur was suppressed using a synchronized light-emitting diode with a 500- μ s pulse duration, substantially shorter than the characteristic timescale of the motion ($\sim 1/f_c > 300$ ms), enabling the capture of instantaneous positions⁴². In each frame, the four elliptical scatterers were fitted with two-dimensional Gaussian functions to extract their centre positions (*x*, *y*) and ellipse parameters (major and minor axis widths). The *z* positions were determined from the scatterer ellipticity using a pre-established calibration curve (Supplementary Fig. 2b). Position tracking precision³⁵, defined as the standard deviation of position measurements for each scatterer in an immobilized micro-platform, was 5 nm along *x*, 7 nm along *y* and 9 nm along *z*. The spatial resolution of the *z*-position determination was validated through controlled displacement experiments: for every 10-nm step along the *z* axis of the immobilized micro-platform, the measured *z* position changed by approximately 10 nm (Supplementary Fig. 3), indicating a spatial resolution better than 10 nm. The translational position of the micro-platform was calculated as the centroid of the four scatterer positions, and its 3D orientation was derived from the scatterer positions using rotation matrices around three axes (a detailed explanation is provided in Supplementary Section 1).

Numerical simulation and theoretical analysis

The 3D electromagnetic simulations of the V-shaped gold nanostructures (the detailed geometry is shown in Fig. 3b) were performed using a finite element solver (COMSOL Multiphysics (v. 6.1)), which has been extensively validated in nanophotonics studies^{11,16,17,19,20,42,53} for its capability to provide accurate, self-consistent solutions to the Maxwell equations. The simulations fully incorporate dissipation effects associated with localized surface plasmons, including both radiative and non-radiative damping, which sensitively depend on the shape and size of the metallic nanostructures. In the simulations, the complex refractive index of gold was implemented according to ref. 62 and the surrounding silicon oxide was modelled with a constant refractive index of $n = 1.45$. To replicate the experimental conditions (Fig. 1b), the simulation modelled a 1×9 array of the V-shaped nanostructures with a 400-nm period, illuminated by circularly polarized Gaussian beams weakly focused to a 1.7- μ m spot diameter.

Optical forces $\mathbf{F}$ and torques $\mathbf{N}$ acting on the nanostructures were calculated from the simulated electromagnetic fields, $\mathbf{E}$ and $\mathbf{H}$, using the Maxwell stress tensor formalism^{31,63}, which is based on the conservation laws of optical linear and angular momenta:

$$\mathbf{F} = \oint_S \overline{\mathbf{T}}_M \cdot \mathbf{n} dS, \quad (6)$$

$$\mathbf{N} = -\oint_S (\overline{\mathbf{T}}_M \times \mathbf{r}) \cdot \mathbf{n} dS. \quad (7)$$

Here the integration is performed over a closed surface S enclosing the nanostructures, $\mathbf{r}$ is the position vector and $\mathbf{n}$ is the outward-pointing unit normal vector. The term $\overline{\mathbf{T}}_M$ denotes the time-averaged Maxwell stress tensor, defined as

$$\overline{\mathbf{T}}_M = \frac{1}{2} \text{Re} \left[\varepsilon_0 \varepsilon \mathbf{E} \otimes \mathbf{E}^* + \mu_0 \mu \mathbf{H} \otimes \mathbf{H}^* - \frac{1}{2} (\varepsilon_0 \varepsilon |\mathbf{E}|^2 + \mu_0 \mu |\mathbf{H}|^2) \mathbf{I} \right], \quad (8)$$

where ε_0 and μ_0 are the permittivity and permeability in a vacuum, respectively; ε and μ are the relative permittivity and permeability of the surrounding medium (silicon oxide in this study) respectively; $\mathbf{I}$ is the unit dyadic; and $\otimes$ denotes the dyadic product operation. In particular, no intrinsic properties of the nanostructures enter the expression for the force and torque; all relevant information is contained in the electromagnetic field on the enclosing surface.

To theoretically analyse the transverse optical torque acting on a plasmonic V-shaped nanostructure, we used equation (2), which was derived from optical forces acting on two coherent dipolar subunits forming the symmetric mode (Fig. 3c) under circularly polarized light illumination¹⁷. Critically, the polarizability α used in equation (2) was rigorously calculated and extracted from the full-wave COMSOL simulations (Supplementary Fig. 11). This self-consistent polarizability of the symmetric mode accounts for both finite-size effects of plasmons and interparticle interactions in metallic nanostructures. By substituting this rigorously calculated polarizability into equation (2), discussed in other work^{23,64,65} for calculating optical forces and torques on arbitrary objects composed of dipolar subunits, the resulting torque showed excellent agreement with that calculated via the Maxwell stress tensor method (Supplementary Fig. 12), thereby confirming the validity of both equation (2) and the polarizability extraction method.

Data availability

Source data are provided with the paper. All other data that support the findings of this study are available from the corresponding author upon reasonable request.

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Author contributions

Y.Y.T. conceived and designed the research project. R.F. and Y.Y.T. performed the sample fabrication. R.F. and Y.Y.T. performed the calculations and established the experimental setup. R.F. and Y.Y.T. analysed the data, with T.S. contributing to the discussions and analysis. R.F. and Y.Y.T. wrote the manuscript with input from all authors.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Yoshito Y. Tanaka.

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